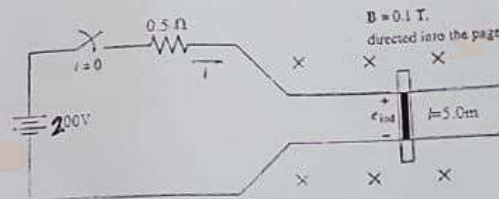




Q2) Solve and show your work

(10 Marks)

11. A single-phase transformer has 1000 turns on the secondary and 100 turns on primary. If the transformer is connected to 110-volt supply while 3 kVA load is connected to the secondary winding. Assuming ideal transformer, the load current is:
a) 2.727 A b) 27.27 A c) 272.7 A d) None of the previous
 12. A single-phase transformer is connected to 110 volt and supply 11 volt with 3 kVA load is connected to the secondary winding. Assuming ideal transformer, the load impedance as referred to the primary is:
a) 0.0004 Ω b) 4.0 Ω c) 4044 Ω d) 403 Ω
 13. A rotor of an electric machine rotates at constant speed and makes 1500 revolutions in three minute. The time required for one complete revolution in seconds is:
a) $\frac{1500}{2\pi}$ b) $\frac{1500}{120\pi}$ c) 0.12 d) 0.002
 14. A transformer with large positive voltage regulation is an indication of
a. Leading load b. lagging load
b. Resistive load d. could be any load
 15. The rated speed of a motor is 1800RPM. If the rated torque of this motor is 40Nm, the rated power of this motor is:
a) 0.5hp b) 5hp c) 10hp d) 100hp
- Q3) For the non-loaded linear DC machine shown in the figure below with voltage source 200 V, choose the correct answer for the following five questions.



16. The starting current of this machine is:
a) 100A b) 300A c) 400A d) 200A
17. The steady state velocity is:
a) 10 m/s b) 20 m/s c) 400 m/s d) 200 m/s
18. If a 20N force is opposing the motion of the machine, the steady-state current consumed by the machine is:
a) 1.0A b) 4.0A c) 40A d) 200A
19. If a 20N force is opposing the motion of the machine, the steady-state velocity of the machine is:
a) 160.0m/s b) 120.0m/s c) 100m/s d) 360 m/s
20. If a 20N force is opposing the motion of the machine, the power consumed by the machine at steady-state is:
a) 1600W b) 3200W c) 4800W d) 7200W

Q1 (1 point)

1	2	3	4	5	6	7	8	9	10

Q2 (2 points)

11	12	13	14	15

Q3 (2 points)

16	17	18	19	20



Q1) Choose the Correct Answer

- The copper of transformer must have
 - low reluctance.
 - high reluctance
 - low resistance.
 - None of the above
- Which of the following losses does not vary with load in transformer?
 - Copper loss.
 - Hysteresis loss.
 - None of the above
 - All losses in transformer vary with load.
- A transformer transform
 - Current
 - Voltage
 - Power
 - Frequency
- The power losses in Real Transformers are _____ ?
 - 5%
 - 25%
 - 50%
 - Different from one transformer to other
 - None of the above
- A single-phase transformer with a ratio of 10:10, if the primary has a torque of 50, then the secondary will generate
 - 50 v
 - 100 v
 - 25 v
 - None of the above
- The electromotive force induced in a conductor of length l , moving with velocity $\vec{V}$ in a magnetic field having flux density $\vec{B}$ is:
 - $e_{ind} = (\vec{V} \times \vec{B}) \cdot \vec{l}$
 - $e_{ind} = (\vec{V} \cdot \vec{B}) \cdot \vec{l}$
 - $e_{ind} = (\vec{V} \times \vec{B}) \cdot \vec{l}$
 - $e_{ind} = (\vec{V} \cdot \vec{B}) \cdot \vec{l}$
- 12) The force exerted on a current-carrying wire placed in a magnetic field is maximum when the angle between the wire and the direction of magnetic field is:
 - 45°
 - 60°
 - 90°
 - 180°
- The law ~~that describes~~ the generated voltage by a varying magnetic field is:
 - a) Gauss's law
 - b) Ampere's law
 - c) Ohm's law
 - d) None of the above
- The primary and secondary windings of transformer are linked each other by
 - eddy current
 - primary induction
 - not linked at all.
 - None of the above
- Which parameter does change during transformation action in transformer?
 - current
 - voltage
 - frequency
 - both A and B
 - None of the above